

Extended Abstract - A novel topological characterization of brain-wide co-fluctuation patterns over time reveals brain-behavior links across spontaneous and evoked activity

Chunyin Siu^a, Saad Pirzada^a, Cameron Glick^a, Jeremy R. Manning^b, Richard Betzel^c, Giovanni Petri^d, Leanne Williams^a, Manish Saggar^a – Corresponding Author

^a Department of Psychiatry and Behavioral Sciences, Stanford University School of Medicine, Stanford, CA, USA

^b Department of Psychological and Brain Sciences, Dartmouth College, Hanover, NH, USA

^c Department of Neuroscience, University of Minnesota, Minneapolis, MN, USA

^d NP Lab, Network Science Institute at Northeastern University London, London, UK

Summary: Traditional analyses of brain functional connectivity using fMRI data rely on Pearson correlation between brain regions: a static, time-averaged measure. While informative, this approach overlooks rich temporal dynamics. Here, we leverage the *edge time series*—the framewise product of fMRI activity between brain region pairs, whose temporal mean recovers the Pearson correlation—to capture fine-grained neural coordination (Faskowitz et al., 2020; Zamani Esfahlani et al., 2020; Owen et al., 2021). To our knowledge, we are the first to apply the Mapper algorithm—a topological data analysis (TDA) method (Singh et al., 2007)—to edge time series, enabling a *visualization of brain-wide co-fluctuation patterns over time*, which clinicians can use directly. This approach fills a key methodological gap in capturing the shape of edge-level dynamics across spontaneous (resting) and task-evoked conditions. We also find that mesoscale properties of the resulting Mapper graphs are significantly associated with individual differences in personality traits and self-reported psychopathology.

Introduction: The brain is a complex network in which spatially distributed regions exhibit coordinated activity over time. Functional connectivity, commonly measured via Pearson correlation between brain region time series, has been widely used to study brain development and psychopathology (Craddock et al., 2013; Rogers et al., 2007). However, as a time-averaged measure, it obscures transient, time-resolved dynamics that may be behaviorally or clinically relevant.

To capture such dynamics, we use the *edge time series*—the moment-by-moment product of activity between each pair of brain regions—whose temporal average yields the traditional Pearson correlation (Faskowitz et al., 2020; Zamani Esfahlani et al., 2020). We then apply the Mapper algorithm (Singh et al., 2007), a topological data analysis (TDA) method, to the edge time series to construct individual-level graphs summarizing the structure of brain-wide co-fluctuation patterns over time.

While previous studies have applied Mapper to node-level fMRI signals (Saggar et al., 2022), our work is the first to apply it to edge-level dynamics, offering a new perspective on the temporal evolution of brain-wide co-fluctuations. Using fMRI data from 265 participants in the Human Connectome Project (Barch et al., 2013; Smith et al., 2004), we generate Mapper graphs for each individual and quantify their mesoscale structure using *modularity*—a graph-theoretic measure that captures the extent to which task states are segregated within the graph (Newman, 2006; Rubinov & Sporns, 2010).

Methods: The input to the Mapper algorithm was the edge time series, computed for each participant using fMRI data concatenated across all seven tasks—working memory, social, relational, motor, language, gambling, and emotion—as well as the resting-state scan. At each time point, the edge time series captures the co-fluctuation between every pair of brain regions, resulting in a high-dimensional temporal profile of brain-wide interactions. For Mapper parameters, we use the correlation metric, we filter by 2D-MDS and we use single-linkage clustering to form nodes.

By concatenating time series across task-evoked and spontaneous (resting) conditions, we aimed to examine whether edge-level dynamics provide a more *characteristic representation* of brain state transitions compared to traditional activation-based (node-level) or higher-order (e.g., triangle-level) signals. This design allowed us to test two key hypotheses: (1) that edge time series more effectively distinguish between task paradigms, with effectiveness measured by the modularity (see Definition 1 below) in the resulting Mapper graphs (partitioned by tasks), and (2) that these topological features are more strongly associated with individual differences in personality traits and self-reported psychopathology. To address this, we generated three distinct Mapper graphs per participant, each corresponding to a different input representation: (1) nodal

activation time series, (2) edge time series (product of brain region pairs), and (3) triangle time series (product of brain region triplets). This allowed direct comparison of how different levels of network representation affect the Mapper-derived topology.

We measured behavioral features in multiple domains, including temperament measured using the NEO Five-Factor Inventory (NEO-FFI) scale (Costa & McCrae, 1992) and internalizing symptoms of depression/anxiety and externalizing symptoms of inattention/impulsivity measured by the Adult Self-Report (ASR; (Achenbach & Edelbrock, 1984; Achenbach & Rescorla, 2003).

Definition 1 (Modularity)

Consider a graph with n_V nodes. Denote by A the adjacency matrix and D_i the degree of node i . For every pair of nodes i and j , let $Q_{ij} = A_{ij} - \frac{D_i D_j}{n_V}$. The modularity Q of the graph with respect to a partition of the nodes is the sum of the Q_{ij} 's for all in-group node pairs i, j (i.e. i and j belong to the same subset in the partition).

Results: To compare how different input representations shape the resulting Mapper graphs, we generated three Mapper graphs for each participant based on input from (1) node-level activation time series, (2) edge time series, and (3) triangle time series. Visual inspection revealed that Mapper graphs constructed from edge time series exhibited greater task-specific segregation (see Fig. 1a–c for a representative participant). In contrast, graphs derived from node and triangle time series showed less clear differentiation between cognitive conditions.

To quantify this segregation, we computed the modularity for each graph, measuring the extent to which task- and rest-related time points were segregated by tasks. As shown in Fig. 1d, modularity peaked for graphs generated from edge time series, compared to both node and triangle inputs. Pairwise t-tests confirmed that these differences were statistically significant across all comparisons (all $p < 0.001$), suggesting that edge-level co-fluctuations best characterize functional brain dynamics across cognitive states.

We next tested whether modularity derived from the edge-based Mapper graphs was associated with individual differences in behavior. We regressed modularity values against seven behavioral features: five NEO-FFI-derived domains of temperament (agreeableness, conscientiousness, extraversion, neuroticism, openness) and two ASR-derived symptom domains (internalizing and externalizing). Before adjusting for age and sex, *conscientiousness* and *externalizing behavior* showed significant associations with modularity ($p < 0.05$). After covariate adjustment, externalizing behavior remained significant ($p = 0.0171$), while conscientiousness approached significance ($p = 0.0525$). Fig. 1e shows the 95% confidence intervals of the regression coefficients for all features after adjustment.

These results support the view that Mapper graphs constructed from edge time series not only capture meaningful cognitive transitions more effectively, but also reflect stable individual differences in personality and psychopathology-relevant behavior.

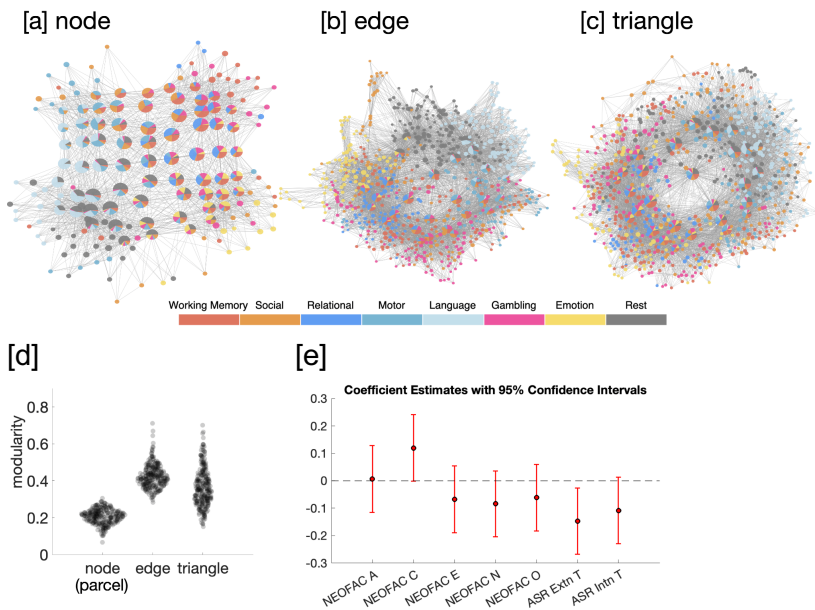


Fig. 1. Mapper graphs for one participant based on (a) node, (b) edge, and (c) triangle time series. Each node represents a cluster of time frames, with pie charts indicating task composition. (d) Swarm plots of modularity values across 265 subjects for each input type. (e) 95% confidence intervals for regression coefficients relating edge-based modularity to behavioral features, adjusted for age and sex.

Conclusion: To conclude, Mapper graphs of edge time series reveal compelling structure in brain activity that warrants further investigation. The observed associations with personality traits and behavioral symptoms suggest that global graph properties may serve as informative markers of individual differences in psychological functioning.

Acknowledgments: This research was supported by an MCHRI Faculty Scholar award and an NIH R01 MH127608 to M.S. C.S. was supported by the Croucher Foundation. Data were provided by the Human Connectome Project, WU-Minn Consortium (Principal Investigators: David Van Essen and Kamil Ugurbil; 1U54MH091657) funded by the 16 NIH Institutes and Centers that support the NIH Blueprint for Neuroscience Research; and by the McDonnell Center for Systems Neuroscience at Washington University.

References

- Achenbach, T. M., & Edelbrock, C. S. (1984). Psychopathology of Childhood. *Annual Review of Psychology*, 35(Volume 35, 1984), 227–256. <https://doi.org/10.1146/annurev.ps.35.020184.001303>
- Achenbach, T. M., & Rescorla, L. A. (2003). *Manual for the ASEBA Adult Forms and Profiles*. University of Vermont, Research Center for Children, Youth and Families, Burlington, VT.
- Barch, D. M., Burgess, G. C., Harms, M. P., Petersen, S. E., Schlaggar, B. L., Corbetta, M., Glasser, M. F., Curtiss, S., Dixit, S., Feldt, C., Nolan, D., Bryant, E., Hartley, T., Footer, O., Bjork, J. M., Poldrack, R., Smith, S., Johansen-Berg, H., Snyder, A. Z., & Van Essen, D. C. (2013). Function in the human connectome: Task-fMRI and individual differences in behavior. *NeuroImage*, 80, 169–189. <https://doi.org/10.1016/j.neuroimage.2013.05.033>
- Costa, P. T., & McCrae, R. R. (1992). *Revised NEO personality inventory (NEO PI-R) and NEP five-factor inventory (NEO-FFI): Professional manual*. Psychological Assessment Resources.
- Craddock, R. C., Jbabdi, S., Yan, C.-G., Vogelstein, J. T., Castellanos, F. X., Di Martino, A., Kelly, C., Heberlein, K., Colcombe, S., & Milham, M. P. (2013). Imaging human connectomes at the macroscale. *Nature Methods*, 10(6), 524–539. <https://doi.org/10.1038/nmeth.2482>
- Faskowitz, J., Esfahlani, F. Z., Jo, Y., Sporns, O., & Betzel, R. F. (2020). Edge-centric functional network representations of human cerebral cortex reveal overlapping system-level architecture. *Nature Neuroscience*, 23(12), 1644–1654. <https://doi.org/10.1038/s41593-020-00719-y>
- Newman, M. E. J. (2006). Modularity and community structure in networks. *Proceedings of the National Academy of Sciences*, 103(23), 8577–8582. <https://doi.org/10.1073/pnas.0601602103>
- Owen, L. L. W., Chang, T. H., & Manning, J. R. (2021). High-level cognition during story listening is reflected in high-order dynamic correlations in neural activity patterns. *Nature Communications*, 12(1), 5728. <https://doi.org/10.1038/s41467-021-25876-x>
- Rogers, B. P., Morgan, V. L., Newton, A. T., & Gore, J. C. (2007). Assessing functional connectivity in the human brain by fMRI. *Magnetic Resonance Imaging*, 25(10), 1347–1357. <https://doi.org/10.1016/j.mri.2007.03.007>
- Rubinov, M., & Sporns, O. (2010). Complex network measures of brain connectivity: Uses and interpretations. *Computational Models of the Brain*, 52(3), 1059–1069. <https://doi.org/10.1016/j.neuroimage.2009.10.003>
- Saggar, M., Shine, J. M., Liégeois, R., Dosenbach, N. U. F., & Fair, D. (2022). Precision dynamical mapping using topological data analysis reveals a hub-like transition state at rest. *Nature Communications*, 13(1), 4791. <https://doi.org/10.1038/s41467-022-32381-2>
- Singh, G., Memoli, F., & Carlsson, G. (2007). Topological Methods for the Analysis of High Dimensional Data Sets and 3D Object Recognition. In M. Botsch, R. Pajarola, B. Chen, & M. Zwicker (Eds.), *Eurographics Symposium on Point-Based Graphics*. The Eurographics Association. <https://doi.org/10.2312/SPBG/SPBG07/091-100>
- Smith, S. M., Jenkinson, M., Woolrich, M. W., Beckmann, C. F., Behrens, T. E. J., Johansen-Berg, H., Bannister, P. R., De Luca, M., Drobnjak, I., Flitney, D. E., Niazy, R. K., Saunders, J., Vickers, J., Zhang, Y., De Stefano, N., Brady, J. M., & Matthews, P. M. (2004). Advances in functional and structural MR image analysis and implementation as FSL. *NeuroImage*, 23, S208–S219. <https://doi.org/10.1016/j.neuroimage.2004.07.051>

Zamani Esfahlani, F., Jo, Y., Faskowitz, J., Byrge, L., Kennedy, D. P., Sporns, O., & Betzel, R. F. (2020). High-amplitude cofluctuations in cortical activity drive functional connectivity. *Proceedings of the National Academy of Sciences*, 117(45), 28393–28401. <https://doi.org/10.1073/pnas.2005531117>